LCA of Temporary Clothing Thrift Store at UMass Lowell

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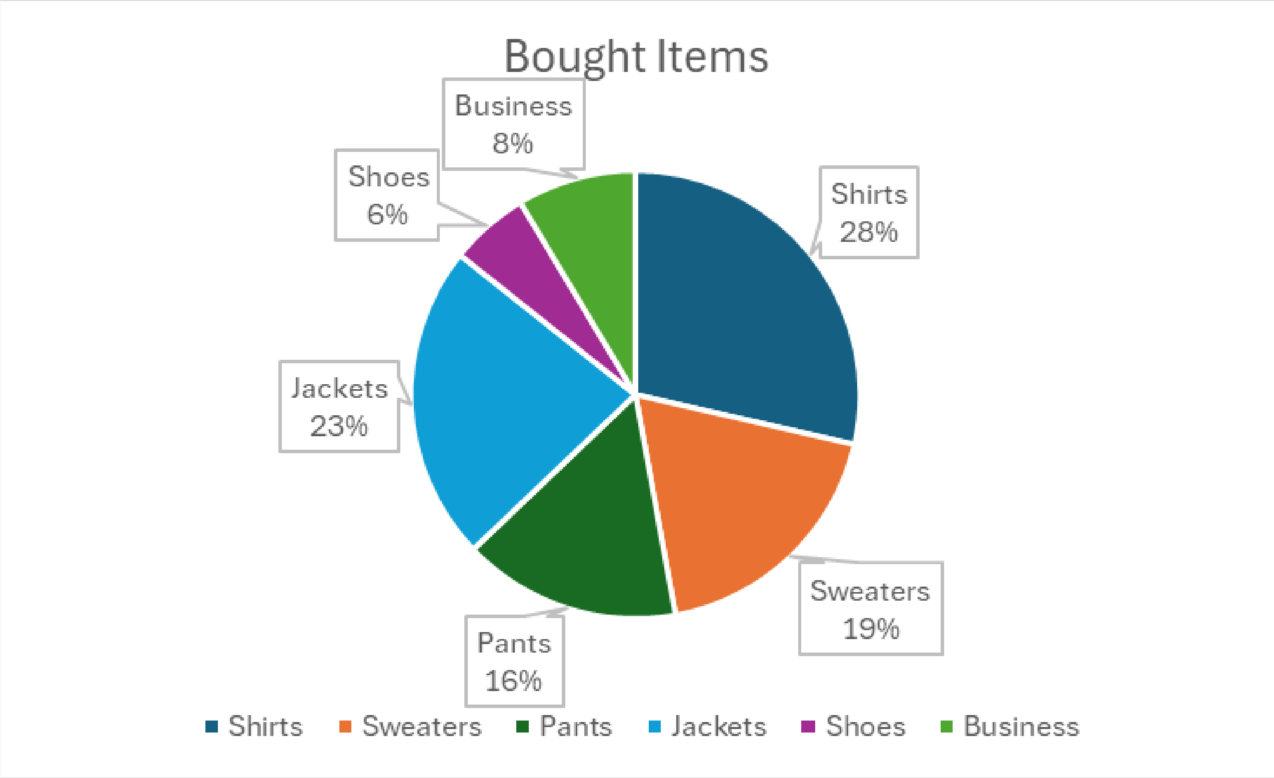
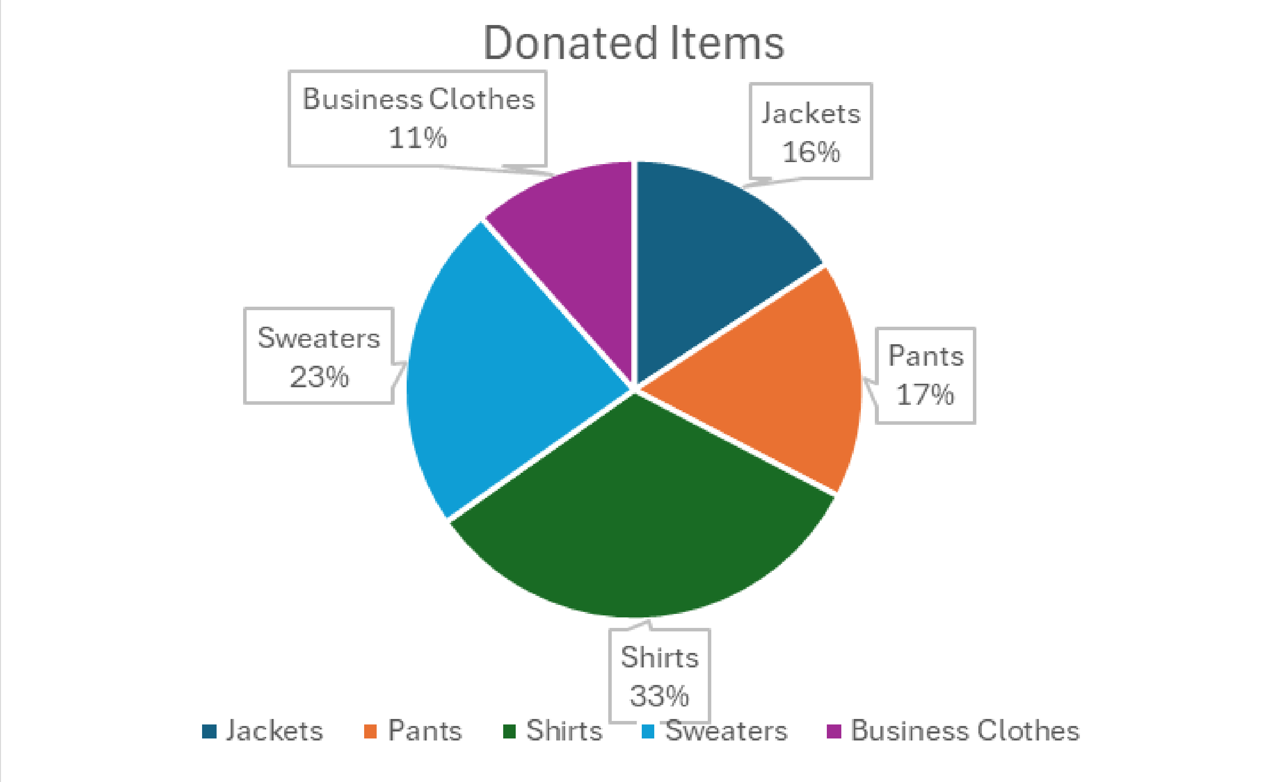
Background and Objectives

- Evaluation of a temporary clothing thrift store on campus that is open for a limited time (4 weeks maximum) per academic year using LCA tools.
  - Energy footprint
  - Environmental footprint
  - Economic cost
- LCA tools:
  - Based on local data such as surveys and data collected from other schools
- Encourage reuse of clothing promoting sustainable habits and student engagement

Recycling Needs – Survey Data



- Would you donate/buy from a UML thrift store?
- What would you donate? What would you buy?



- 87% of those who would donate would also buy clothes
- Out of 100 surveys

Recycling Needs

- Items: Clothing

- Annual flow (in and out):
  - In flow: 0.91 ton/week *(4 weeks of operation) = 3.66 tons per year
  - Out flow: 87%(3.66 tons per year) = 3.18 tons per year
  - Waste flow: 13%(3.66 tons per year) = 0.475 tons per year

Thrift Store Comparisons

UMASS AMHERST

- Began in 2013 led by students
- Diverted 100,000 pounds from going to landfills
- New2U sells thousands of new or gently used items back to students, faculty, and staff at affordable prices on campus
- Partners with Baystate Textile to recycle textiles

NORTHEASTERN

- Pop-up thrift market organized by the Fashion Society
- Secondhand clothing items sold by 13 student vendors.
- Held on campus
- Encourages sustainable fashion practices and provides a platform for student entrepreneurs by promoting handmade pieces to be sold.

Annual Material Flow Estimation

UMass donations	100000	lb
UMass donations	50	ton
Umass per week (30-week academic year)	1.67	ton per wk
UML to Umass Population Ratio	55%	
UML weekly donations	0.91	ton per wk
UML waste (13% not bought)	0.12	ton per wk
Total donated per year	3.66	ton per year
Total waste per year	0.475	ton waste per year

Chosen Model

- LCA objective: minimize cost and minimize emissions
- Duration: 4 weeks per academic year of Donations, 2 weeks of Selling Clothes
  - Week Leading up to Thanksgiving Break (Selling)
  - Week Leading up to Winter Break
  - Week Leading up to Spring Break (Selling)
  - Week Leading up to Summer Break/Move-Out
- Disposal Method:
  - Landfill & WTE Plant
- System Boundaries:
  - UML (for Donations)
  - UML to LF/WTE (waste disposal)

Store and Storage

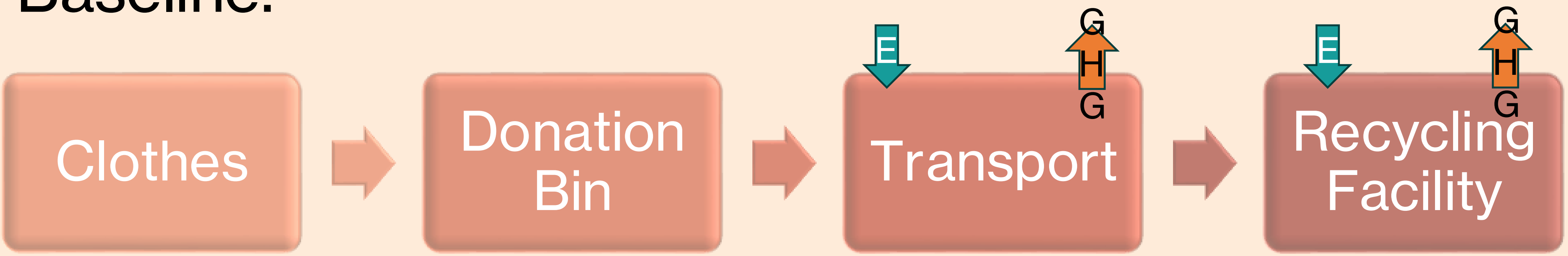
- With 80% of storage used up a room roughly 101 square ft would be needed to store the clothing.
- Thrift Store can be held at UCrossing (building about 230000 sf) or outside if weather permits
- Storage closets in back first floor of UCrossing if permission is granted or Jimmy B lounge storage closet.
- Thrift shop labor can be structured through volunteers and the hours they can do, proposing ~6 hours per day
- Proposed Actions for Student Awareness: Social media (pre and post engagement), Emails, Flyers around campus, Possible incentives handed out by volunteers, Collaborate with clubs (sustainability, fashion, etc.)
- Sorting, inventory, and individual pricing would not occur everything would be thrown into a pile which would allow for less labor

Store Operation Estimation

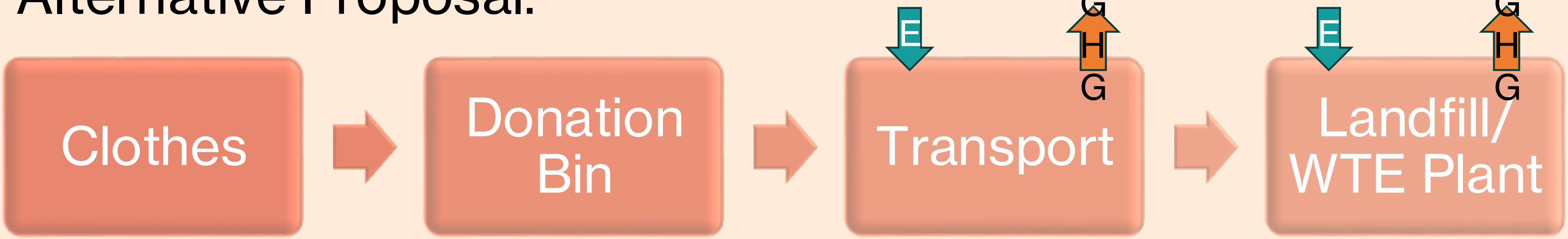
- Pricing:
  - $100/ton
  - Incentive Deal: 50% off per pound if you donate
- Vehicle Type to Transport Waste:
  - Small Recyclable Items in Passenger Car (Can fit up to 550lb of clothing)
  - To bring donations: hire workers to move donation bins to UCrossing (no energy required)
- Chosen Landfill:
  - Pelham Transfer Station (8 mi from UML)
- Chosen Waste-to-Energy Incineration Plant:
  - Wheelabrator North Andover (29 mi from UML)

Annual Material Flow

Baseline:



Alternative Proposal:



Life-Cycle Analysis

Material	Process	Unit Qty (ton)	Energy Footprint (electric) (MJ)	Energy Footprint (diesel) (MJ)	GHG footprint (kg CO2 eq)
Mixed Textile	Landfill	1	2.88	38.6	8310
Mixed Textile	Incineration	1	0	225	5900
Mixed Textile	MRF	1	22.5	0	8.58
Small Recyclable Items	Transport by Car to LF	0.275	0	1014.66	41.6
Small Recyclable Items	Transport by Car to WTE Plant	0.275	0	3678.14	150.8
Small Recyclable Items	Transport by Car to MRF	0.275	0	9030.47	370.24
Transport by Car	50 MJ/ ton/mi				
Distance to LF		8mi	Pelham Transfer Station		
Distance to Inc		29mi	Wheelabrator North Andover		
Distance to MRF		71.2mi	Bay State Textiles		

	Baseline: Municipal Recycling Facility			
	Transport	Total E Consumption:	4303	MJ
		GHG Emission:	180	kg CO2 eq
	Alternative: 2 Options			
Landfill	Total E Consumption:		502	MJ
		GHG Emission:	3969	kg CO2 eq
Waste to Energy	Total E Consumption:		1855	MJ
		GHG Emission:	2876	kg CO2 eq

Life-Cycle Analysis (cont.)

Cost Analysis

- Transport energy: 50 MJ/ton/mi
- Vehicle operation cost: ~$0.60/mi
- Labor cost: **$25/hour**, average trip duration estimated at 1.5 hours round-trip

Transport by Car	50 MJ/ton/mi			
Distance to LF	8	mi	Pelham Transfer Station	
Distance to Inc	29	mi	Wheelabrator North Andover	
Distance to MRF	71.2	mi	Bay State Textiles	

Cost Comparison (per 0.275 Tons)

Labor	$25/hour			
Total Cost Analysis				
Method	Labor Cost	Transport Cost	Processing Cost	Total Cost
Landfill	$ 37.50	$ 1.32	$ 24.75	$ 63.57
WTE	$ 37.50	$ 4.79	$ 19.25	$ 61.54
MRF	$ 37.50	$ 11.74	$ 13.75	$ 62.99
Thrift Shop*	$ -	$ -	$ - (27.50)	$ - (27.50)

*$0 or even **- $100/ton** (if resale generates revenue)

Annual Financial Impact Using UML Waste Rate

Annual Impact Using UML Waste Rate			
Avoided cost (if landfill was baseline):			
0.475 x $63.57=	$ 30.28		
Thrift Shop Revenue Opportunity:			
0.475 x $100=	$ 47.50		
Net Benefit:	**$ 77.78**	**per year**	

Conclusion

- Viable option from LCA and cost perspectives
- Proposed reuse strategy based on findings: Temporary Thrift store with an annual WTE trip for waste is best
- The proposed thrift store viability is checked through:
  - Cost: $34.04 per 0.275 tons (best)
  - LCA:
    - Energy consumption:1855 MJ
    - GHG Emissions: 2876 kg CO_2 eq

Resources

- https://www.umass.edu/gateway/umass-edge/about-umass-amherst/umass-amherst-numbers
- https://www.sciencedirect.com/science/article/abs/pii/S0956053X23003525#:~:text=The%20study%20reported%20that%20MRFs,et%20al.%2C%202022
- https://www.sciencedirect.com/science/article/abs/pii/S0956053X14004346#:~:text=MRF%20electricity%20consumption%20ranges%20from,to%20%2424.9%20per%20Mg%20input.
- https://www.eia.gov/Energyexplained/units-and-calculators/
- https://www.umass.edu/sustainability/new2u
- https://www.baystatetextiles.com/
- https://huntnewsnu.com/81312/lifestyle/northeastern-fashion-society-brings-student-thrift-market-to-campus/